

INTERNATIONAL ENVIRONMENTAL LAW AND GLOBAL ISSUES

A. INTRODUCTION

In international law, a distinction is often made between hard and soft law. Hard international law generally refers to agreements or principles that are directly enforceable by a national or international body. Soft international law refers to agreements or principles that are meant to influence individual nations to respect certain norms or incorporate them into national law. Although these agreements sometimes oblige countries to adopt implementing legislation, they are not usually enforceable on their own in a court.

If a treaty or convention does not specify an international forum that has subject matter jurisdiction, often the only place to bring a suit with respect to that treaty is in the member state's domestic court system. This presents at least two additional hurdles. If the member state being sued does not have domestic implementing legislation in place to hear the dispute, there will be no forum available. Even in the event that the domestic legislation provides for such suits, since the judges who decide the case are residents of the country against which it is brought, potential conflicts of interest arise.

Only nations are bound by treaties and conventions. In international forums, such as the International Court of Justice (ICJ), countries must consent to being sued. Thus, it is often impossible to sue a country. The final question in the jurisdictional arena is who may bring a suit. Often, only countries may sue countries. Individual citizens and non-governmental organizations (NGOs) cannot. This has huge repercussions. First, the environmental harm must be large and notorious for a country to notice. Second, for a country to have a stake in the outcome of the subject matter, some harm may have to cross the borders of the violating country into the country that is suing. Finally, even if transboundary harm does exist, the issue of causation,

especially in the environmental field, is often impossible to prove with any certainty.

The enforcement issue is one where advocates for a safer environment often find themselves stymied. Even if a treaty or convention provides for specific substantive measures to be taken by a country (many treaties merely provide 'frameworks'), specifies a forum for dispute resolution and authorizes sanctions for non-compliance, international law remains largely unenforceable. A country cannot be forced to do what it is not willing to do. One can sanction the country, order damages, restrict trade, or, most frequently, publicize non-compliance. But beyond that, if a country will not comply, there is very little to be done.

International institutions are generally not responsible for directly implementing and enforcing international environmental law, but they often play important monitoring, informational and diplomatic roles. For example, the 1992 Convention on the Conservation of Biological Diversity (Biodiversity Convention)¹ created a new international body, the Committee on Sustainable Development (CSD). The CSD lacks the power to bring enforcement actions against either governments or private parties, but it plays a role in implementing the Biodiversity Convention. The CSD helps monitor national compliance efforts by requiring member nations to submit annual reports. Through its meetings and publications, the CSD also provides a forum to discuss and debate issues associated with global protection of biological diversity and forests.

NOTES AND QUESTIONS

1. Consider for a moment why any law is enacted—domestically or internationally. Some would maintain that it is a moral statement about behaviour that a society cannot tolerate. Some would argue that certain conduct is outlawed to deter that conduct, which is why we also attach a penalty. Some would argue, especially in light of the inefficiencies in enforcement, that laws socialize society's members to behave in a certain way by defining a code.

What is the purpose of international environmental law—is it a moral statement, a deterrence, or a socializing tool? If it is a moral statement, which many of the framework conventions seem to be, is it merely aspirational? Do we honestly believe that all nations will achieve all the ideals expressed in all the agreements? Or do we, as a global community, simply like to think of ourselves as the kind of people who believe in these things? If it is intended as deterrence, why are there not more international forums for dispute resolution, more international bodies empowered to enforce agreements, more substantive requirements, and more 'hard law' self-executing agreements? If there were, would any nation sign them? If it is intended as a socialization technique, is it working? Are nations more environmentally aware?

¹ Reprinted in 31 I.L.M. 818 (1992) and P. Birnie and A. Boyle, *Basic Documents on International Law and Environment*, 390 (1995).

2. If ultimately all international environmental law is unenforceable, what good is it? Does it accomplish anything to find a country out of compliance with a treaty? What about publicity? What if the economic benefits of a project such as the Narmada Valley Project, are believed by government officials to outweigh the negative effects of the publicity?

3. The practice of relying on domestic implementing legislation to enforce international environmental agreements leave state parties in the position of having different obligations under the same treaty, depending on how their legislative, executive and judicial bodies interpret and implement the treaty. Is this fair? What about the costs and administrative burdens that are associated with creating and enforcing legislation? Does this put richer countries in a better position to comply with treaties? Does not this mean that poorer countries, who may wish to comply, will often be unable to? Should they be held to the same standards? Is it patronizing to hold them to *lower* standards?

4. What is the purpose of the informational roles of international institutions? Will more knowledge about the global environment and our impacts on it lead to better compliance? Or will so many new issues lead to non-compliance due to uncertainty? If it appears to the average citizen that virtually everything she does has a negative environmental impact, will she not cease to try to change any behaviour?

5. What are the difficulties with the monitoring role of international institutions? If information gathered by monitoring bodies is generated by the governmental institutions, how accurate may it be? Is it better to have inaccurate information or no information? How much of a threat to a nation's sovereign status is an international monitoring institution? When does a nation's sovereignty have to be compromised for the greater global good? Who is to say?

B. INDIA'S INTERNATIONAL OBLIGATIONS

India has obligations under numerous international treaties and agreements that relate to environmental issues. As a contracting party, India must have ratified a treaty, that is, by adopting it as national law before it came into force, or by acceding to it after it has come into force. For a treaty to enter into force, the requisite number of countries must ratify the treaty, which then has the force of international law.

Specific obligations under any treaty vary, depending on the treaty itself. The nature and degree of compliance and implementation depend on a number of factors, among them: (1) the capabilities and staff of an international institution charged with coordinating national compliance efforts; (2) the willingness of other state parties to enforce or comply with the treaty; (3) the political agenda of the government and popular support; (4) trade and diplomatic pressures brought to bear by other countries; and (5) sometimes, judicial or NGO involvement through court cases and publicity.

INDIA'S TREATY OBLIGATIONS

1. The Antarctic Treaty (Washington, 1959) 402 UNTS 71. Entered into force 23 June 1961. India ratified with qualifications, 19 August 1983.
2. Convention on Wetlands of International Importance, Especially as Waterfowl Habitat (Ramsar, 1971). 11 I.L.M. 963 (1972). Entered into force 21 December 1975. India acceded, 1 October 1981.
3. Convention Concerning the Protection of the World Cultural and Natural Heritage (Paris, 1972). 11 I.L.M. 1358 (1972). Entered into force 17 December 1975. India signed, 16 November 1972.
4. Convention on International Trade in Endangered Species of Wild Fauna and Flora (Washington, 1973) 12 I.L.M. 1055 (1973). Entered into force 1 July 1975. India signed, 9 July 1974; ratified 20 July 1976.
5. Protocol of 1978 Relating to the International Convention for the Prevention of Pollution from Ships, 1973 (MARPOL) (London, 1978). Entered into force 2 October 1983. India ratified with qualifications, 24 September 1986.
6. Convention on the Conservation of Migratory Species of Wild Animals (Bonn, 1979) 19 I.L.M. 15 (1980). Entered into force 1 November 1983. India signed, 23 June 1979; ratified 4 May 1982.
7. Convention on the Conservation of Antarctic Marine Living Resources (Canberra, 1980). 19 I.L.M. 841 (1980). Entered into force 7 April 1982. India ratified, 17 June 1985.
8. United Nations Convention on the Law of the Sea (Montego Bay, 1982). 21 I.L.M. 1261 (1982). Entered into force 16 November 1994. India signed, 10 December 1982.
9. Convention for the Protection of the Ozone Layer (Vienna, 1985). 26 I.L.M. 1529 (1987). Entered into force 22 September 1988. India ratified, 18 March 1991.
10. Protocol on Substances That Deplete the Ozone Layer (Montreal, 1987). 26 I.L.M. 1550 (1987). Entered into force 1 January 1989. India acceded, 19 June 1992.
11. Amendments to the Montreal Protocol on Substances That Deplete the Ozone Layer (London, 1990). 30 I.L.M. 541 (1991). Entered into force 10 August 1992. India acceded, 19 June 1992.
12. Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal (Basel, 1989). 28 I.L.M. 657 (1989). Entered into force 5 May 1992. India signed, 5 March 1990; ratified 24 June 1992.
13. United Nations Framework Convention on Climate Change (Rio de Janeiro, 1992). 31 I.L.M. 849 (1992). Entered into force 21 March 1994. India signed, 10 June 1992; ratified 1 November 1993.
14. Convention on Biological Diversity (Rio de Janeiro, 1992). 31 I.L.M. 818 (1992). Entered into force 29 December 1993. India signed, 5 June 1992; ratified 18 February 1994.

15. Convention to Combat Desertification in Those Countries Experiencing Serious Drought and/or Desertification, Particularly in Africa (Paris, 1994). 33 I.L.M. 1332 (1994). Entered into force, 26 December 1995; India signed, 14 October 1994; ratified 17 December 1996.
16. International Tropical Timber Agreement (Geneva, 1994). 33 I.L.M. 1016 (1994). Entered into force 1 January 1997. India signed, 17 September 1996. India ratified 17 October 1996.
17. Protocol on Environmental Protection to the Antarctica Treaty (Madrid, 1991). Entered into force 15 January 1998.

C. ESTABLISHED NORMS OF INTERNATIONAL ENVIRONMENTAL LAW

Norms are general legal principles that are widely accepted. This acceptance is evidenced in a number of ways, such as international agreements, national legislation, domestic and international judicial decisions, and scholarly writings. The leading norms in the field of international environmental law are addressed below:

(1) Foremost among these norms is Principle 21 of the 1972 Stockholm Declaration on the Human Environment. Principle 21 maintains that 'States have, in accordance with the Charter of the United Nations and the principles of international law, the *sovereign right to exploit their own resources* pursuant to their own environmental policies, and *the responsibility to ensure that activities within their jurisdiction or control do not cause damage to the environment of other States or of areas beyond the limits of national jurisdiction*'.²

(2) Another widely shared norm is the *duty of a state to notify and consult* with other states when it undertakes an operation that is likely to harm neighbouring countries' environments, such as the construction of a power plant, which may impair air or water quality in downwind or downstream states.

(3) Over and above the duty to notify and consult, a relatively new norm has emerged whereby states are expected to *monitor and assess* specific environmental conditions domestically, and disclose these conditions in a *report* to an international agency or international executive body created by an international agreement, and authorised by the parties to the agreement to collect and publicize such information.

(4) Another emerging norm is the guarantee in the domestic constitutions, laws or executive pronouncements of several states, including India,³ Malaysia, Thailand, Indonesia, Singapore and the Philippines, that all *citizens have a right to a decent and healthful environment*. In the United States, this fundamental

² Emphasis supplied. Reprinted in P. Birnie and A. Boyle, *id.* at 1.

³ The fundamental right to life guaranteed under Article 21 of the Indian Constitution has been interpreted by the Supreme Court to include the right to a wholesome environment. *Subhash Kumar v State of Bihar* AIR 1991 SC 420, 424.

right has been guaranteed by a handful of states but not by the federal government.

(5) Most industrialized countries subscribe to the *polluter pays principle*. This means polluters should internalize the costs of their pollution, control it at its source, and pay for its effects, including remedial or cleanup costs, rather than forcing other states or future generations to bear such costs. This principle has been recognized by the Indian Supreme Court as a 'universal' rule to be applied to domestic polluters as well.⁴ Moreover, it has been accepted as a fundamental objective of government policy to abate pollution.⁵

(6) Another new norm of international environment law is the *precautionary principle*. This is basically a duty to foresee and assess environmental risks, to warn potential victims of such risks and to behave in ways that prevent or mitigate such risks. In the context of municipal law, Justice Kuldeep Singh of the Supreme Court has explained the meaning of this principle in the *Vellore Citizens' Welfare Forum Case*,⁶ which is excerpted later in this section.

(7) *Environmental impact assessment* is another widely accepted norm of international environmental law. Typically, such an assessment balances economic benefits with environmental costs. The logic of such an assessment dictates that before a project is undertaken, its economic benefits must substantially exceed its environmental costs. India has adopted this norm for select projects which are covered under the Environmental Impact Assessment (EIA) regulations introduced in January, 1994.⁷

(8) Another recent norm is *to invite the input of non-governmental organizations (NGOs)*, especially those representing community-based grassroots environmental activists. This NGOs participation ensures that the people who are likely to be most directly affected by environmental accords will have a major role in monitoring and otherwise implementing the accord. This principle is mirrored in the Indian government's domestic pollution control policy⁸ and the national conservation policy,⁹ and is given statutory recognition in the EIA regulations of 1994. The Supreme Court has urged the government to draw upon the resources of NGOs to prevent environmental degradation.¹⁰

(9) In October 1982, the United Nations General Assembly adopted the World

⁴ *The Bichhri Case (Indian Council for Enviro-Legal Action v Union of India)* AIR 1996 SC 1446; and *Vellore Citizens' Welfare Forum v Union of India* AIR 1996 SC 2715.

⁵ Ministry of Environment and Forests, Government of India, *Policy Statement for Abatement of Pollution*, para 3.3 (26 February 1992).

⁶ *Supra* note 4. In *A.P. Pollution Control Board v Prof. M.V. Nayudu* AIR 1999 SC 812 the Supreme Court traced the development of the precautionary principle.

⁷ The EIA regulations are extensively dealt with in the Large Projects chapter.

⁸ *Supra* note 5, at para 11.1.

⁹ Ministry of Environment and Forests, Government of India, *National Conservation Strategy and Policy Statement on Environment and Development*, para 8.7 (June, 1992).

¹⁰ *Indian Council for Enviro-Legal Action v Union of India (CRZ Notification Case)* 1996 (5) SCC 281.

Charter for Nature and Principles of *Sustainable Development*. The agreement expressly recognised the principle of sustainable development, defined as using living resources in a manner that 'does not exceed their natural capacity for regeneration' and using 'natural resources in a manner which ensures the preservation of the species and ecosystems for the benefit of future generations.' The principle of sustainable development was also acknowledged in the 1987 report *Our Common Future*, published by the United Nations World Commission on Environment and Development. This report defined sustainable development as 'humanity's ability ... to ensure that [development] meets the need of the present generation without compromising the ability of future generations to meet their needs.' The Supreme Court¹¹ as well as the Indian government have recognised the principle of sustainable development as a basis for balancing ecological imperatives with developmental goals.¹²

(10) *Intergenerational equity* is among the newest norms of international environmental law. It can best be understood not so much as a principle, but rather as an argument in favour of sustainable economic development and natural resource use. If present generations continue to consume and deplete resources at unsustainable rates, future generations will suffer the environmental (and economic) consequences. It is our children and grandchildren who will be left without forests (and their carbon retention capacities), without vital and productive agricultural land and without water suitable for drinking or for sustaining cultivation or aquatic life. Therefore, we must all undertake to pass on to future generations an environment as intact as the one we inherited from the previous generation.

Proponents of intergenerational equity maintain that the present generation has a moral obligation to manage the earth in a manner that will not jeopardize the aesthetic and economic welfare of the generations that follow. From this moral premise flow certain ecological commandments: 'Do not cut down trees faster than they grow back. Do not farm land at levels, or in a manner, that reduce the land's regenerative capacity. Do not pollute water at levels that exceed its natural purification capacity.'

In *State of Himachal Pradesh v Ganesh Wood Products*¹³ the Supreme Court recognized the significance of inter-generational equity and held a government department's approval to establish forest-based industry to be invalid because 'it is contrary to public interest involved in preserving forest wealth, maintenance of environment and ecology and considerations of sustainable growth and inter-generational equity. After all, the present generation has no right to deplete all the existing forests and leave nothing for the next and future generations.'¹⁴

(11) At the 1982 United Nations Conference on the Law of the Sea

¹¹ *Vellore Citizens' Welfare Forum v Union of India* AIR 1996 SC 2715.

¹² *Supra* note 9, at paras 1.1, 1.3.

¹³ AIR 1996 SC 149, 163. Also see *CRZ Notification Case*, *supra* note 10, where the court expressed its concern at the adverse ecological effects which will have to be borne by future generations.

¹⁴ *Id.*

(UNCLOS),¹⁵ developing countries, led by India, articulated the norm that certain resources, such as the deep seabed, are part of the *common heritage of mankind* and must be shared by all nations.

(12) The 1992 Rio de Janeiro Earth Summit articulated the norm of *common but different responsibilities*. With regard to global environmental concerns such as global climate change or stratospheric ozone layer depletion, all nations have a shared responsibility, but richer nations are better able than poorer nations to take the financial and technological measures necessary to shoulder the responsibility.

NOTES AND QUESTIONS

1. As mentioned, norms of customary international law evolve through custom and usage. Not all norms are of equal importance however, some being accorded the status of *fundamental norms*. The category of fundamental norms comes under the doctrine of *jus cogens*, or the doctrine of *peremptory norms*. The 1969 Vienna Convention on the Law of Treaties¹⁶ serves to clarify the concept in Article 53 as follows:

A treaty is void if, at the time of its conclusion, it conflicts with a peremptory norm of general international law. For the purposes of the present Convention, a peremptory norm of general international law is a norm accepted and recognised by the international community of States as a whole as a norm from which no derogation is permitted and which can be modified only by a subsequent norm of general international law having the same character.

The doctrine of *jus cogens* is extremely limited and extends to only a handful of norms, the most long-standing of which are the prohibitions against the slave trade, piracy and genocide. Many scholars also believe that the norm expressed in Principle 21 of the Stockholm Convention has risen to *jus cogens* status. Principle 21 is based on the Roman maxim, *sic utero tuo et alienum non laedas*, which roughly means 'do not behave in a way that hurts your neighbour.' However, again the question arises, what good does an international peremptory norm, like an international agreement, achieve?

2. Consider some of the other norms addressed above, such as the right to a healthful environment. Is there to be one standard by which all environments are judged, or is it a relative concept? If cutting down trees for firewood destroys the environment, but provides life-sustaining fuel, which right will prevail? Which right *should* prevail? How much weight does each of the two rights carry when the concept of intergenerational equity is introduced?

¹⁵ Reprinted in 21 I.L.M. 1261 (1982).

¹⁶ Reprinted in 8 I.L.M. 679 (1969).

When considering moral ideals such as the principles of a common heritage and intergenerational equity, what incentives do countries have to try to mould their practices to achieve these ideals? How can countries be better motivated? Is saving the environment for its own sake going to appeal to the majority of people, or does there have to be a more direct benefit, like the idea that we may, by destroying an ecosystem, inadvertently destroy the cure for cancer?

3. What exactly is 'sustainable development'? What is 'sustainable'? Could not intelligent and informed people differ over whether producing more minerals or preserving a landscape is sustainable? If so, which is more important? And what is 'development'? Factories that employ thousands, may give a country more industry with which to compete in the world market, but discharge effluents into the water and produce piles of hazardous waste. Do some countries have more of a right to development and less of an obligation to ensure sustainability and vice versa? How much more do industrialised nations need to develop? Have they not gone far enough? Who is to judge?

At the end of a judicial career which earned him the sobriquet 'Green Judge', Justice Kuldip Singh of the Supreme Court issued comprehensive directions to clean up the mess created by the leather tanneries of Tamil Nadu. In the following excerpt from this leading case, Justice Singh borrows international law norms and applies them to the local milieu.

VELLORE CITIZENS' WELFARE FORUM v UNION OF INDIA

AIR 1996 SC 2715

KULDIP SINGH, J.:

* * *

The traditional concept that development and ecology are opposed to each other, is no longer acceptable. 'Sustainable Development' is the answer. In the International sphere 'Sustainable Development' as a concept came to be known for the first time in the Stockholm Declaration of 1972. Thereafter, in 1987 the concept was given a definite shape by the World Commission on Environment and Development in its report called 'Our Common Future'. The Commission was chaired by the then Prime Minister of Norway Ms.G.H.Brundtland and as such the report is popularly known as 'Brundtland Report'. In 1991 the World Conservation Union, United Nations Environment Programme and World Wide Fund for Nature, jointly came out with a document called 'Caring for the Earth' which is a strategy for sustainable living. Finally, came the Earth Summit held in June, 1992 at Rio which saw the largest gathering of world leaders ever in the history—deliberating and chalking out a blue print for the survival of the planet. Among the tangible achievements of the Rio

Conference was the signing of two conventions, one on biological diversity and another on climate change. These conventions were signed by 153 nations. The delegates also approved by consensus three non-binding documents namely, a Statement on Forestry Principles, a declaration of principles on environmental policy and development initiatives and Agenda 21, a programme of action into the next century in areas like poverty, population and pollution. During the two decades from Stockholm to Rio 'Sustainable Development' has come to be accepted as a viable concept to eradicate poverty and improve the quality of human life while living within the carrying capacity of the supporting eco-systems. 'Sustainable Development' as defined by the Brundtland Report means 'development that meets the needs of the present without compromising the ability of the future generations to meet their own needs'. We have no hesitation in holding that 'Sustainable Development' as a balancing concept between ecology and development has been accepted as a part of the Customary International Law though its salient features have yet to be finalized by the International Law Jurists.

Some of the salient principles of 'Sustainable Development', culled-out from Brundtland Report and other international documents, are inter-generational equity; use and conservation of natural resources; environmental protection; the precautionary principle; polluter pays principle; obligation to assist and cooperate; eradication of poverty and financial assistance to the developing countries. We are, however, of the view that 'the precautionary principle' and 'the polluter pays' principle are essential features of 'Sustainable Development'. The 'precautionary principle'—in the context of the municipal law—means:

- (i) Environmental measures—by the State Government and the statutory authorities—must anticipate, prevent and attack the causes of environmental degradation.
- (ii) Where there are threats of serious and irreversible damage, lack of scientific certainty should not be used as a reason for postponing measures to prevent environmental degradation.
- (iii) The 'Onus of proof' is on the actor or the developer/industrialist to show that his action is environmentally benign.

'The polluter pays' principle has been held to be a sound principle by this Court in *Indian Council for Enviro-Legal Action vs. Union of India (The Bichhri Case)*, 1996 (3) SCC 212. The Court observed, 'We are of the opinion that any principle evolved in this behalf should be simple, practical and suited to the conditions obtaining in this country'. The Court ruled that 'Once the activity carried on is hazardous or inherently dangerous, the person carrying on such activity is liable to make good the loss caused to any other person by his activity irrespective of the fact whether he took reasonable care while carrying on his activity. The rule is premised upon the very nature of the activity carried on'. Consequently the polluting industries are 'Absolutely liable to compensate for the harm caused by them to villagers in the affected area, to the soil and to the underground water and hence, they are bound to take all necessary measures to remove sludge and other pollutants lying in the affected areas'. The 'polluter pays' principle as interpreted by this Court means that the absolute liability for harm to the environment extends not only to compensate the victims of pollution but also the cost of restoring the environmental degradation. [Remedying] the damaged environment is

part of the process of 'Sustainable Development' and as such [the] polluter is liable to pay the cost to the individual [who] suffers as well as the cost of reversing the damaged ecology.

[The court then set out the provision of the Constitution as well as the Water Act, Air Act and Environment (Protection) Act]. In view of the above mentioned constitutional and statutory provisions we have no hesitation in holding that the precautionary principle and the polluter pays principle are part of the environmental law of the country.

Even otherwise once these principles are accepted as part of the Customary International Law there would be no difficulty in accepting them as part of the domestic law. It is almost accepted proposition of law that the rule of Customary International Law which are not contrary to the municipal law shall be deemed to have been incorporated in the domestic law and shall be followed by the Courts of Law. For support we may refer to Justice H.R.Khanna's opinion in *Addl. Distt. Magistrate Jabalpur vs. Shivakant Shukla* (AIR 1976 SC 1207), *Jolly George Verghese's case* (AIR 1980 SC 470) and *Gramophone Company's case* (AIR 1984 SC 667).

The Constitutional and statutory provisions protect a person's right to fresh air, clean water and pollution free environment, but the source of the right is the inalienable common law right of clean environment. [The court proceeded to quote a paragraph from Blackstone's commentaries on the Laws of England (1876) in respect of 'nuisance']. Our legal system having been founded on the British Common Law the right of a person to pollution free environment is a part of the basic jurisprudence of the land.

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NOTES AND QUESTIONS

1. In the concluding part of the *Vellore* judgment (which has not been reproduced above), the Supreme Court issued a wide range of directions. The Central Government was required to establish an authority under section 3(3) of the Environment (Protection) Act of 1986 headed by a retired High Court judge, to deal with the situation created by the tanneries and other polluting industries in Tamil Nadu. The authority so constituted was asked to apply the precautionary principle and the polluter pays principle whilst identifying the individuals who had suffered due to the pollution, assessing the loss incurred and determining the compensation payable by the polluters. The authority was conferred wide powers by the court, including the power to direct the closure of industry and the relocation of units. The court also imposed a pollution fine of Rs.10,000 on each of the tanneries in five districts of the state.

2. The discussion in the judgment relating to the polluter pays principle is dissatisfactory. In *Indian Council for Enviro-Legal Action (The Bichhri Case)* the

Supreme Court recognized the distinction between the principle of 'absolute liability' applicable only to hazardous industries, and the wider 'polluter pays' principle which applies whenever an industry pollutes the air, water or land, irrespective whether it is a hazardous unit or not. The court in the *Vellore Case* appears to have mixed up these two distinct principles and rolled them into one. Would you interpret the Vellore judgment to mean that the 'no-fault' liability principle extends to all situations where the damage is caused by pollution, and is no longer restricted in India to damage caused by hazardous industries? Does this mean that in India a polluter is no longer entitled to the defences that are available to him under the law of torts?

D. CONFLICTS BETWEEN THE ENVIRONMENT AND FREE TRADE

To help promote responsible environmental practices at home and abroad, many countries have enacted legislation that contains certain trade restrictions that prevent the free flow of goods across borders. Such environmental restrictions often conflict with the terms of international free trade agreements, primarily with the General Agreement on Tariffs and Trade (GATT), which seeks to discourage or prohibit the use of import restrictions.

GATT was formed in 1947 in the aftermath of World War II, primarily to encourage global economic development by limiting the use of tariffs and import restrictions. The agreement is the initial 1947 document and its periodic 'rounds' of amendment. The World Trade Organization (WTO) is the administrative body that sits in Geneva, Switzerland whose purpose is to implement the terms and requirements of the GATT. One way that WTO achieves this end is through dispute resolution panels where member states can reconcile conflicting interpretations of GATT's provisions.

GATT allows nations to restrict the import of products from other member nations, so long as these restrictions do not discriminate between foreign and domestic products. For instance, under GATT the US may ban the importation of a dangerous pesticide so long as the use of the pesticide is also banned in the US. Article 20 lists the exceptions that justify a deviation from GATT's general free trade requirements. Among these exceptions are trade restrictions 'necessary to protect human, animal or plant life and health,'¹⁷ and those 'relating to the conservation of exhaustible natural resources.'¹⁸ Most of the controversy regarding the agreement's impact on environmental protection has centered on the WTO dispute panels' interpretations of the Article 20 exceptions. No trade restriction based on Article 20(b) or 20(g) has as yet been upheld by these dispute panels.

¹⁷ Article 20 (b).

¹⁸ Article 20 (g).

NOTES AND QUESTIONS

1. GATT is one of the most powerful international agreements in the world and often supersedes environmental treaties. Why is this? Is it because GATT, unlike most environmental treaties, has an institutional dispute resolution mechanism and enforcement procedures? Or is it because the global community puts greater emphasis on commodities and trade than on the environment and conservation?

2. Many treaties, recognizing the power of market-based incentives, have tried to turn environmental issues into commodity issues. Is the market the proper mechanism for saving the environment? Would not the unfettered operation of free trade, as pictured under GATT, lead to the eventual destruction of the environment? Has it not brought us perilously close to that point already? Why, then, are economic approaches and GATT so powerful? Is economic self-interest the only true motivator of people and nations? If so, is there a way to give greater economic value to the environment? Is there an economic theory or accounting mechanism that can take true environmental costs into account in the face of a project's touted economic benefits?

E. HUMAN RIGHTS, INDIGENOUS PEOPLE AND THE ENVIRONMENT

Although environmental protection and human rights are often treated as separate legal topics, there are many situations where the two fields overlap. First, many governments and international bodies have recognised the right of citizens to live in a clean and healthful environment. Second, environmental and natural resource policies may disproportionately affect poor and minority communities.

The rights of indigenous people are a 'cross-over' issue in that they may be protected under both international human rights law and international environmental law. (Any environmental right can theoretically be couched in terms of a human right, so this is not the only area where the two bodies of law dovetail.)

The rights of indigenous people may be seen in two basic lights: (1) the right to protect and manage natural resources located on traditional indigenous lands; (2) the right of citizens to live in a healthful environment. Many environmentally destructive development practices severely impact the traditional lands and lifestyles of indigenous communities. Therefore, their rights often provide another tool in the fight against such projects.

With respect to the protection of indigenous people as a means to conserve biodiversity, international environmental law can play an important role. Many native and indigenous people have opposed government policies that permit resource exploitation on traditional lands. Because this exploitation threatens to undermine the economic and spiritual fabric of their cultures, and often results in forced migration and resettlement, the struggle to protect the environment is often a part of the struggle to protect the cultures of indigenous people.

THE RIO DECLARATION ON ENVIRONMENT AND DEVELOPMENT

Out of the Earth Summit in 1992, three documents emerged, one of which was the Rio Declaration on Environment and Development.¹⁹ In Principle 22, the Declaration states that '[i]ndigenous people and their communities and other local communities have a vital role in environmental management and development because of their knowledge and traditional practices. States should recognize and duly support their identity, culture and interests and enable their effective participation in the achievement of sustainable development.'

The same principle is echoed in the Biodiversity Convention²⁰ in Article 8(j). However, as discussed below in the section on Protecting Endangered Species, both the Declaration and the Convention lack substantive obligations and enforcement mechanisms. Thus, both are regarded as 'framework' conventions upon which truly binding rights and responsibilities can only be built through further agreements.

The combination of human rights and environmental obligations can provide a powerful tool to protect land and ecosystems where indigenous people reside. This protection is evidenced by the reservation of lands for indigenous peoples, such as the Yanomani lands in Brazil, that incidentally also contain some of the world's precious rainforests. As the conservation expertise of indigenous peoples becomes more widely acknowledged, indigenous rights are likely to play an increased role in national and international environmental protection efforts.

NOTES AND QUESTIONS

1. Why do indigenous people deserve special protection? Granted, they have traditionally been dispossessed and marginalized, but so have many other groups, such as women. What value do indigenous people add to the global community? Is 'saving' indigenous people akin to saving rainforests? Should we do it because they are 'there' and it is therefore morally wrong to destroy them? In preserving biodiversity, how much *human* diversity should we aim to preserve? Do we only protect indigenous people because there is 'something in it' (medicinal and agricultural knowledge) for the rest of us? Is it altruism, selfishness or guilt that motivates us?

2. What if a community that suffers from an environmental threat or hazard is *not* indigenous? Do they therefore deserve less protection? What if indigenous people are inhabiting fragile lands that ought to remain pristine? What if indigenous people are engaged in environmentally destructive practices? When does the protection of indigenous people, and their common property lands, conflict with their right to development and self-determination?

¹⁹ Reprinted in P.Birnie and A. Boyle, *supra* note 1, at 9. The Supreme Court referred to Principle 13 of the Declaration requiring states to develop national laws to compensate the victims of pollution and other environmental damage in the *CRZ Notification Case*, *supra* note 10.

²⁰ *Supra* note 1.

F. ENVIRONMENTAL JUSTICE

I. ENVIRONMENTAL POVERTY LAW

There is a growing awareness of the close relationship between poverty and environmental pollution. It is broadly recognized that poorer citizens are more likely to suffer the consequences of environmental pollution than other citizens, on both the national and international levels. It has also given rise to environmental poverty law, or environmental justice, which seeks legal remedies to the disproportionate environmental abuse suffered by poorer citizens.

Internationally, less affluent nations tend to have more severe environmental problems than wealthier nations. Examples of these problems are easy to identify. Air pollution in Mexico and China is generally more severe than in France or Australia. Hazardous waste is treated less safely in Eastern Europe and Africa than in Western Europe and Canada. The reasons for this situation are frustrating but not difficult to understand. Less affluent nations lack the financial resources to purchase modern pollution control or energy efficient technologies, or to implement environmental protection policies.

At the 1992 Earth Summit in Rio, developing countries asked for increased technology transfers. These countries pointed out that if the developed world is truly concerned with stemming the environmental deterioration in developing countries, new technologies for environmental protection need to be made available at little or no cost to the developing world.

On the national level, there have also been important developments. In India, for example, judicial decisions have held that the urban poor must be treated fairly and that government policies must respect their human rights.²¹ Similarly, the United States has also begun to address environmental poverty issues. In 1994, President Clinton issued an executive order calling on federal agencies to make certain that environmentally undesirable activities do not disproportionately burden certain economic or ethnic sectors of society.

II. NORTH-SOUTH TENSIONS

Tensions between the developed and developing world are found at the international level. Many transnational corporations headquartered in the developed world have chosen to move environmentally dangerous industrial activities to the developing world.

Why do transnational corporations relocate to these poorer nations? First, labour costs are much cheaper. Second, taxes are substantially lower. Finally, environmental standards in the developing world are generally lower (and therefore less expensive) than in the developed world. This means, of course, that workers are often exposed to hazardous materials or unsafe conditions.

²¹ e.g., *Olga Tellis v Bombay Municipal Corporation* AIR 1986 SC 180.

The export of environmental harm to the world's poorer nations, and to non-European populations, has resulted in severe health and environmental problems. Two widely-publicized examples were the lethal 1984 Union Carbide gas leak in Bhopal, and the continuing logging of tropical rainforests by First World timber companies.

NOTES AND QUESTIONS

1. Environmental justice has become one of the 1990s most sensitive issues. It underlies almost all of the negotiating sessions and the disputes leading up to international environmental agreements. Poorer, less developed countries (LDCs) feel strongly that developed countries exercise 'eco-imperialism' in all arenas. Is this correct? Why? Does it help powerful nations stay powerful? By not transferring technology, do the rich get richer? At whose expense? Is it not in the best interest of the developed countries, who after all share the planet with the LDCs, to ensure that LDCs have the technologies and economic resources to invest in sustainable development projects and industries, in order to preserve the global commons (the deep sea, the atmosphere, Antarctica)? Keep in mind that global environmental degradation, like global climate change and stratospheric ozone depletion, recognizes no boundaries.

2. The United States' initial refusal under President Bush to sign the 1992 Convention on Biological Diversity (discussed below) was based on one of the most controversial environmental justice issues, namely intellectual property rights over plant genetic resources. The privatization of biotechnology and plant genetic resources, primarily derived from LDCs but patented in developed countries, has led to wide-scale protests and civil unrest, resulting in what has become known as 'seed wars'. Are not seeds, as well as all plant genetic resources, humankind's 'common heritage'? Should they be allowed to be patented? If so, should they be patented in their countries of origin? Would allowing countries of origin to patent, thereby obtaining royalties, provide an economic incentive to conserve them?

G. AIR POLLUTION AND PROTECTION OF THE ATMOSPHERE

During the 1980s, international environmental efforts shifted from transboundary or regional air pollution concerns, such as acid rain, to threats to the global atmosphere. This shift was prompted by scientific evidence that emerged in the mid 1970s. The evidence linked the release of chlorofluorocarbons (CFCs) and other chlorine-based substances with the destruction of the stratospheric ozone layer. The ozone layer shields human beings and plant life from the harmful effects of solar radiation. CFCs are used in refrigeration, air conditioning, and foam furniture among other applications. To reduce the use of these substances, and protect the global atmosphere, the 1985 Vienna Convention for the Protection of

the Ozone Layer²² and the 1987 Montreal Protocol on Substances that Deplete the Ozone Layer²³ were adopted. Because the 1987 protocol incorporated the 1985 convention's basic aims, the international regime to protect the ozone layer is usually referred to as the Montreal Protocol.

Another issue involving the world's atmosphere is global climate change. The issue of global climate change encompasses many underlying environmental concerns, including air pollution, energy consumption, deforestation and management of the global commons. For purposes of this chapter, the issue will be discussed in the present section on air pollution and protection of the atmosphere. The primary international agreements relating to global warming and climate changes are the United Nations Framework Convention of Climate Change (UNFCCC)²⁴ signed at the Rio de Janeiro Earth Summit in 1992, and the Kyoto Protocol to the UNFCCC, signed at Kyoto, Japan in 1997.

1987 MONTREAL PROTOCOL ON SUBSTANCES THAT DEplete THE OZONE LAYER

The 1987 Montreal Protocol on Substances that Deplete the Ozone Layer (Montreal Protocol) sets firm targets for reducing consumption and production of a range of ozone-depleting substances. The standards set forth in the 1987 Protocol were strengthened and expanded to cover additional ozone-depleting substances, through amendments adopted in 1990, 1992 and 1994. These amendments were prompted by the development of new technology and alternative 'ozone friendly' substances. The Protocol's ability to respond to and incorporate scientific/technological developments has been widely praised, and has provided a model of constructive flexibility for future international environmental agreements.

One of the major innovations of the Montreal Protocol is its recognition that all nations should not be treated equally. The agreement acknowledges that certain countries have contributed greatly to ozone depletion while other countries have made very small contributions. The agreement also recognizes that a nation's obligation to reduce current emissions should reflect its technological and financial ability to abate CFC pollution. Because of this situation, the agreement applies more stringent standards, and a more accelerated phase-out timetable, to the countries that have contributed the most to ozone depletion. Such countries must eliminate all CFCs by the year 2000. Developing countries have a 10 year 'grace period', and must phase out CFC use by 2010.

The Montreal Protocol also includes innovative funding provisions providing less affluent member countries financial and technical incentives (such as the transfer of technology and patents) to encourage a rapid switch to non-ozone depleting substances and production methods. Specifically, Article 10 of the Protocol established a fund to facilitate technical co-operation and technology transfer to assist developing states.

²² Reprinted in 26 I.L.M. 1529 (1987) and P. Birnie and A. Boyle, *supra* note 1, at 211.

²³ Reprinted in 26 I.L.M. 1550 (1987) and P. Birnie and A. Boyle, *id.* at 224.

²⁴ Reprinted in 31 I.L.M. 849 (1992) and P. Birnie and A. Boyle, *id.* at 252.

This Multilateral Ozone Fund, now administered by the Global Environment Facility (GEF),²⁵ depends on the support of the developed countries. While this system seems fair, it also creates a loophole in the Protocol through which many less developed countries (LDCs) will be able to avoid meeting the standards set out in the Protocol if the developed countries do not meet their financial obligations.

Finally, the Protocol also contains provisions to deal with the problem of the few nations that have not signed the Protocol and continue to produce and consume ozone-depleting substances, by banning trade in these substances with non-member states. Thus, parties to the Protocol are prohibited from importing such substances, or exporting CFC production technology and equipment. This comprehensive trade ban places both economic and diplomatic pressure on all nations to join the Protocol.

1992 UNITED NATIONS FRAMEWORK CONVENTION ON CLIMATE CHANGE

The Climate Change Convention²⁶ was prompted by several scientific studies in the late 1980s which indicated that increased levels of carbon dioxide (CO₂) in the atmosphere were causing global temperatures to rise. This increase in temperatures was attributed to the 'greenhouse effect,' wherein the sun's heat is trapped between the earth's surface and the CO₂ in the earth's atmosphere (which prevents the heat from escaping). The Climate Change Convention was adopted to reduce the amount of CO₂ emitted into the atmosphere, and to preserve and increase the earth's carbon absorption capacities.

In addressing the global warming issue, the international community chose to follow the process successfully employed in the ozone/CFC context. Just as the general, aspirational 1985 Vienna Convention preceded the highly specific 1987 Montreal Protocol, the 1992 Framework Convention represents the first step in the international community's attempt to stop global warming. Its purpose is to demonstrate and forge consensus, and to provide the diplomatic foundation for a more substantive agreement.

Article 2 of the Convention states that the 'ultimate objective' of the framework agreement is to stabilize the concentrations of greenhouse gases at a level which would prevent dangerous interference with the climate system. This broad and general phrasing provides participating countries with flexibility in implementation strategies.

The Climate Change Convention was only a broad blueprint, but some significant principles and provisions were negotiated. Most of these provisions reflect North-South (developed-developing world) tensions. First, it was agreed that financial commitments should be based on the principles of 'respective capabilities', and 'appropriate burden sharing' and 'equity', meaning that wealthier nations should be required to contribute more than poorer nations. Second, the Convention states that

²⁵ GEF was initially established in 1991 as a pilot programme of the World Bank, UNEP and the UN Development Programme. It was restructured in 1994 following the decisions taken at the Rio Conference. The 1994 instrument is *reprinted* in 33 I.L.M. 1273 (1994).

²⁶ *Reprinted* in 31 I.L.M. 849 (1992) and P.Birnie and A. Boyle, *supra* note 1, at 252.

developed countries 'shall take all practical steps to promote, facilitate and finance, as appropriate, the transfer of, or access to, environmentally sound technologies.' In addition, the Framework Convention specifies that the newly created Global Environmental Facility will act as the financial mechanism for allocating environmental resources to developing countries.

Although the Climate Change Convention is modelled on the Montreal Protocol/CFC approach, the recent meetings at Kyoto (1997) and Buenos Aires (1998) show that it may be difficult to move from the aspirational 'framework' stage to the binding 'implementation' stage. This is because the global warming issue lacks many of the elements that formed the foundation for the Montreal Protocol/CFC regime. Most importantly, there is lack of available and affordable alternatives to thermal power, and all nations, North and South, contribute to the problem and must contribute to its solution.

In December, 1997, the Third Conference of Parties to the Framework Convention on Climate Change met in Kyoto, Japan. After much contentious dispute and negotiation, the parties agreed and signed the Kyoto Protocol. The major industrialized countries agreed to reduce their greenhouse gas emissions by an average of five per cent relative to their 1990 levels, in the period 2008 to 2012, Japan agreed to a reduction of six per cent, the US agreed to seven per cent, and the European Union agreed to eight per cent. (Most countries, including the US, have yet to ratify the Protocol.)

Developing nations rejected taking on any new commitments and only agreed at the last minute to allow emissions trading among all countries as part of a 'clean development mechanism'. India and China seem prepared to resist emissions trading schemes because they want developed nations to bear the major costs of global carbon reductions—even though such reductions may be achieved in energy-inefficient developing countries at much lower cost than in energy-efficient developed countries. The 'clean development mechanism' of Article 12 of the Protocol seemingly allows developed countries to implement (and pay for) carbon-reducing projects in developing countries. This provision is largely undefined.

NOTES AND QUESTIONS

1. Having gained the experience of moving from framework to substantive agreements in the case of the Montreal Protocol, why did the drafters of the Climate Change Convention seek to *repeat* the framework process? Now that we know it can be done, why not move directly to the implementing protocol and skip the framework step altogether? Are there formalities involved in international negotiations that must be observed? Is this an efficient way to solve environmental problems? What if the problems are extremely urgent?

2. Since virtually all climate scientists agree on the likelihood of global climate change, why has collective action been so slow in coming? What happened to the

precautionary principle discussed above? How accurate and reliable does data have to be before we act on it? What if the effects of waiting cause irreparable harm?

3. What about GEF involvement in these projects? GEF is sponsored, in part, by the World Bank; the two institutions share staff and are located in the same site. Is this a problem? If so, how can it be overcome?

H. PROTECTING ENDANGERED SPECIES

1973 CONVENTION ON INTERNATIONAL TRADE IN ENDANGERED SPECIES OF WILD FAUNA AND FLORA

As of 1995, over 113 nations had signed the 1973 Convention on International Trade in Endangered Species (CITES).²⁷ CITES does not seek to directly protect endangered species or the development practices that destroy their habitats. Rather, it seeks to reduce the economic incentive to kill endangered species and destroy their habitat by closing off the international market. CITES' sole aim is to control or prevent international commercial trade in endangered species or products derived from such species.

CITES regulates by means of an international permit system. For plant and animal species threatened with extinction, international import or export is strictly forbidden. For plant and animal species suffering decline but not yet facing extinction, international import/export permits must be secured. These CITES permits enable the trade to be controlled and monitored so that it does not lead to species extinction or decline.

Many environmentalists believe that CITES does not go far enough in protecting endangered species, and that it approaches protection from a philosophically questionable standpoint. The main animals that CITES protects are 'charismatic species'—so called megafauna or animals whose parts and pelts have a market value. This prioritising of species is objectionable to many. The only thing that CITES controls is trade; it does nothing to limit hunting or killing. Theoretically, if a nation wanted to kill all its species for domestic consumption, CITES could do nothing to prevent it. Many people believe that CITES' dependence on trade regulations to protect endangered species is inimical to a biocentric perspective.

1992 BIODIVERSITY CONVENTION

At the 1992 UNCED meeting in Rio de Janeiro, an agreement was reached on the conservation and sustainable use of the world's biodiversity. The Convention on Biological Diversity (Biodiversity Convention)²⁸ took effect on 29 December 1993, after it was ratified by the required minimum of 30 countries.

²⁷ Reprinted in 12 I.L.M. 1055 (1973) and P.Birnie and A.Boyle, *id.* at 415.

²⁸ *Supra* note 1.

Although the Biodiversity Convention sets forth numerous obligations, most of these are aspirational; there are no specific standards or methods to ensure compliance. Article 8(c) requires that signatory nations 'regulate or manage biological resources important for the conservation of biological diversity whether within or outside protected areas, with a view to ensuring their conservation and sustainable use.' Article 8(d) obliges countries to 'promote the protection of ecosystems, natural habitats and the maintenance of viable populations of species in natural surroundings.' Under Article 9(b), nations agree to 'adopt measures relating to the use of biological resources to avoid or minimise adverse impacts on biological diversity.'

The environmentally progressive provisions listed above, however, are balanced against provisions that reaffirm each nation's sovereign right to manage and exploit its natural resources. Countries may adopt nature protection standards that are appropriate to their own economic needs and priorities. Taken as a whole, the Biodiversity Convention is broad and vague enough to be consistent with almost all natural resource policies, whether these policies are environmentally protective or destructive. The loose and contradictory language of the agreement has been criticized by many environmentalists, who maintain that an important opportunity was missed to create more sustainable international nature protection standards.

The Convention has also been criticized for its lack of enforcement mechanisms. The Convention created an institution, the Committee on Sustainable Development (CSD), to monitor compliance and implement the agreement. The CSD, however, has so far been unable to establish precise definitions of the vague and contradictory language in the agreement. Without this basic clarification, it is difficult to implement the Convention or monitor compliance.

NOTES AND QUESTIONS

1. If Kenya decided to kill every elephant in its borders for purposes of internal domestic consumption, CITES would not prevent this policy from being carried out. What kind of environmental treaty is it, then? Is it not actually another commodities agreement in disguise? CITES is often mentioned as the environmental treaty most likely to be voidable under GATT. Why is this so?

2. Is it true, as many critics contend, that CITES is only good for the animals we humans find appealing? Do we not have a moral obligation to protect *all* animals? Or if we only conserve out of utilitarian motivations, all *useful* animals? Of course, we can't possibly know which animals will prove useful!

3. The Biodiversity Convention is regarded by many as an unsuccessful attempt to please all of the people all of the time. Can all its conservation principles truly coexist with its sovereignty principles? In fact, can inviolable sovereignty coexist with global environmental law in general?

4. While the Biodiversity Convention has been criticized for being only a framework, is it possible to create a treaty with more specific substantive obligations

covering a topic as large as the planet's entire biodiversity? Is it more manageable to attempt to conserve different ecosystems individually, rather than impose guidelines that try to apply to all types in one instrument?

I. PROTECTING THE GLOBAL COMMONS

There are some areas of sea, land, and air that do not fall within recognized national boundaries. These areas are referred to as the 'global commons.' A primary example of global commons is international waters at sea—waters located outside each country's 200 mile exclusive economic zone.

1982 UNITED NATIONS CONVENTION ON THE LAW OF THE SEA

The United Nations Law of the Sea Convention (UNCLOS)²⁹ established several duties regarding the marine environment. These obligations include the duty of nations to (1) 'protect and preserve the marine environment'; (2) 'take, individually or jointly as appropriate, all measures that are necessary to prevent, reduce and control pollution of the marine environment from any source, using for this purpose the best practical means at their disposal and in accordance with their capabilities'; (3) 'take all measures necessary to ensure that activities under their jurisdiction or control are so conducted as not to cause damage by pollution.'

UNCLOS also obliges nations to (1) cooperate on a global and regional basis with international organizations to formulate 'international rules, standards and recommended practices and procedures for the protection and preservation of the marine environment'; (2) 'cooperate in the promotion of scientific research and data exchange programme regarding marine pollution'; (3) cooperate 'in eliminating the effects of pollution and preventing or minimizing the damage'; and (4) establish appropriate scientific criteria for the formulation of international environmental 'rules, standards and recommended practices and procedures for the prevention, reduction and control of marine pollution.'

Implementation and enforcement of UNCLOS is left to individual member nations. One national strategy to ensure compliance is to prohibit dumping or destructive fishing techniques by all vessels registered in a particular country. Another strategy is to deny port privileges to vessels from nations that have failed to adopt national legislation implementing UNCLOS. Both of these strategies are means of controlling activity in international waters located outside national boundaries.

A major legacy of the Law of the Sea treaty process is the assertion by the so-called group of 77 developing nations that the deep seabed, and minerals contained there, are the 'common heritage of mankind.' It logically follows that the profits from any mining of the deep seabed must be shared with all countries. Thus far, it has not

²⁹ *Supra* note 15

proved cost effective to mine the deep seabed. Also, the United States has not accepted the 'common heritage of mankind' principle as applied to the deep seabed. Obviously, this is an area of international environmental law that is still evolving.

NOTES AND QUESTIONS

1. Why would the United States not accept the UNCLOS 'common heritage of mankind' principle? Is it because the US, as a richer, technically advanced nation, may be one of the first to mine the deep seabed?

2. One of the most difficult aspects of marine issues is the monitoring of behaviour on the open sea. How would any country know if another country observed the principles of international law in the middle of an ocean? Does this render all future international agreements with respect to oceans largely aspirational?

3. Given the difficult regulatory nature of the open sea, would it not make more sense to mandate design changes in ships, such as double hulling to insure against oil spills, that could be enforced in the port states or in the building process?

J. PROTECTING FOREST ECOSYSTEMS

1992 STATEMENT OF FOREST PRINCIPLES

At the 1992 Rio de Janeiro Conference, more than 178 states adopted a statement of principles for the sustainable management of forests. Although the principles adopted are non-binding and lack the force of international law, the statement does provide a good basis upon which a future legally-binding multilateral agreement may be built.

The agreement title, *Forest Principles: Non-Legally Binding Authoritative Statement of Principles for a Global Consensus on the Management, Conservation and Sustainable Development of all Types of Forests*,³⁰ reflects the difficult politics that surrounded its drafting and adoption. The Forest Principles statement calls for information and technological exchange among parties to the agreement. It encourages public participation, including that of indigenous peoples likely to be affected by a proposed forest project, acknowledges the extreme importance of sustainable forestry practices, and specifically states that 'special attention should also be given to the countries undergoing the process of transition to market economics.'

From the history of the development of the Forest Principles statement, it is evident that some international support does exist for a binding multilateral forest agreement. During the annual 'Group of Seven Industrial Nations' (G-7) economic summit in 1990, the G-7 heads of government endorsed negotiation of a forest protection treaty. Since the G-7 economic summit, the United States has favoured a

³⁰ Reprinted in 31 I.L.M. 881 (1992).